

8-31 ALTAM Additional practice solutions

1. (10 points) A life insurer issues a 5-year term insurance, with premiums payable half-yearly, to Korina, who is age 60. The death benefit of 100,000 is paid at the end of the half-year of death.

You are given:

- (i) Expenses are 10% of each semi-annual premium.
- (ii) $i = 0.05$
- (iii) Mortality follows the Standard Ultimate Life Table; the force of mortality of the Standard Ultimate Life Table is:
 $\mu_{x+t} = A + Be^{ct}$, where $A = 2.2 \times 10^{-4}$, $B = 2.7 \times 10^{-6}$, $c = 1.124$
- (iv) The gross premium is 270 per half-year.

- (a) (3 points)
- (i) Show that $100,000 {}_{0.5}q_{64} = 260$ to the nearest 10. You should calculate the value to the nearest 1.
 - (ii) Show that the gross premium reserve at time 4 is 30 to the nearest 10. You should calculate the reserve to the nearest 0.1.
- AS3501/AS5102
Questions
Try these!

a) (i) $0.5q_{64} = 1 - 0.5p_{64}$

The formula for Makeham ${}_t p_x$ is given in the formula sheet.

$$0.5p_{64} = e^{(-0.00022(0.5) - \frac{0.0000027}{\ln 1.124} 1.124^{64} (1.124^{0.5} - 1))}$$

$$= e^{-0.002577} = 0.997427$$

$$100,000 {}_{0.5}q_{64} = 100,000 (1 - 0.997427)$$

$$= 257.35$$

Comments: Candidates generally did well in this part and most of them received full credit. Mistakes include:

- Using force of mortality as a constant force of mortality; and
- Using constant force of mortality assumption and the survival probability from the table for Part c while assuming uniform distribution of deaths.

(ii) ${}_4V = \text{APV of future benefits} + \text{APV of future expenses}$
 $\quad - \text{APV of future premiums}$

$$= 100,000 (v^{0.5} (0.002574) + v (0.997427) (0.5q_{64.5}))$$

$$+ (0.1(270) + 0.1(270)(0.997427)v^{0.5})$$

$$- (270 + 270(0.997427)v^{0.5})$$

$$= 30.14$$

Comments: Generally, most candidates knew to use either EPV or recursive formula to compute the reserve. Common errors made by candidates include:

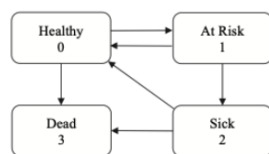
- Not properly handling the semi-annual premiums;
- Approximating semi-annual annuity using annual annuity; and
- Assuming uniform distribution of death or constant force of mortality.

$$* 0.5p_{64.5} = e^{(-0.00022(0.5) - \frac{0.0000027}{\ln 1.124} 1.124^{64.5} (1.124^{0.5} - 1))}$$

$$= e^{-0.002725} = 0.997278$$

$$0.5q_{64.5} = 0.002722 *$$

In the final year of the policy, the insurer decides to re-evaluate the gross premium reserve due to a virus that is spreading through the community. The insurer uses the following multiple state model.



1. Continued

A Healthy person will move to State 1, At Risk, if exposed to the virus. Every person moving into the At Risk state will be tested for the virus. If the result of the test is positive, the person moves to State 2, Sick; otherwise, the person will move back to the Healthy state.

The transition intensities are:

$$\mu_x^{01} = 0.06$$

μ_x^{03} is the force of mortality of the Standard Ultimate Life Table, as in (iii) above.

$$\mu_x^{10} = 20.0; \quad \mu_x^{12} = 80.0$$

$$\mu_x^{20} = 4.0; \quad \mu_x^{23} = 5\mu_x^{03}$$

You are also given that Korina is Healthy at time 4, and that premiums are payable conditional on being in State 0, 1, or 2.

- ✓ (b) (2 points) Calculate the probability that Korina will remain in the Healthy state throughout the final year of the policy, ${}_1P_{64}^{00}$.

our class

$$\begin{aligned}
 {}_1P_{64}^{00} &= e^{-\int_0^1 (\mu_{x+t}^{01} + \mu_{x+t}^{03}) dt} \\
 &= e^{-\int_0^1 (0.06 + \mu_{x+t}^{03}) dt} \\
 &= e^{-0.06(1)} \cdot e^{-\int_0^1 \mu_{x+t}^{03} dt}
 \end{aligned}$$

μ^{03} is from the SULT
so we can use P_{64} from the table

$$= (0.941765)(1 - 0.005288)$$

$$= \underline{0.936784}$$

Comments: Most candidates received full credit for this part.